

THE SUPER-YANGIAN SHADOW TOWER: RICCATI EXPANSION AND THE $\kappa_{\text{ch}} + \kappa_{\text{ch}}^! = \max(m, n)$ COMPLEMENTARITY

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ABSTRACT. For super-Yangian inputs $Y(\mathfrak{sl}(m|n))$, the Sugawara-shifted Verdier dual satisfies

$$\kappa_{\text{ch}}(Y(\mathfrak{sl}(m|n))) + \kappa_{\text{ch}}(Y(\mathfrak{sl}(n|m))^!) = \max(m, n).$$

The zero-sum analogy is false for this normalization. The super-Riccati expansion gives closed shadow coefficients S_r whose characteristic arithmetic primes differ from the bosonic Virasoro tower. The super-trace and Berezinian pairings give two normalizations of the same complementarity datum. The canonical Verdier pairing remains an explicit open construction.

PRIMARY THEOREMS

- $\kappa_{\text{ch}}(Y(\mathfrak{sl}(m|n))) + \kappa_{\text{ch}}(Y(\mathfrak{sl}(n|m))^!) = \max(m, n)$: (symbolically verified at small rank; replaces naive = 0 analogy).
- Super-Riccati closed forms for S_r through $r = 8$: .
- Canonical Verdier pairing (super-trace vs Berezinian canonicalisation): (open construction; the Verdier pairing is not yet inscribed).

CHAPTER INPUTS (POINTERS)

```
\input{../chapters/examples/super_riccati_shadow_tower_platonic.tex}
\input{../chapters/theory/quantum_groups_foundations.tex}
\input{../chapters/theory/quantum_chiral_algebras.tex}
```

CROSS-VOLUME DEPENDENCIES

- Vol I `riccati.tex` standalone (bosonic Riccati algebraicity and depth classification).
- Vol I shadow-tower higher-coefficients theorem (`thm:shadow-series-closed-form-Virasoro`).
- Vol II tempering bound β_N .

CORRECTIONS

The super-Yangian complementarity identity has right-hand side $\max(m, n)$ in the Sugawara-shifted dual normalization. The Virasoro-style constant belongs to a distinct family. The tempered stratum has $\limsup \kappa_{\text{ch}}^{(\infty)}_{\text{orig}} = 0$ at generic c ; the value $1/e$ comes from a Stirling cancellation error.